International Journal of Multidisciplinary Research and Growth Evaluation.

Deep Learning with LLM: A New Paradigm for Financial Market Prediction and Analysis

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Article Info

ISSN (online): 2582-7138
Volume: 06
Issue: 02
March-April 2025
Received: 11-01-2025
Accepted: 15-02-2025
Page No: 211-215

Abstract
This study presents the development of an AI text generation detection tool based on the Transformer model, aiming to enhance the accuracy of AI text generation detection and provide a reference for future research. The tool employs a series of preprocessing steps on the text, including Unicode normalization, conversion to lowercase, removal of non-alphabetic characters using regular expressions, and the addition of spaces around punctuation marks. It also eliminates leading and trailing spaces, replaces consecutive ellipses with a single space, and connects text segments with a specified delimiter. Subsequently, non-alphabetic characters and redundant spaces are removed, and multiple consecutive spaces are replaced with a single space, followed by a second conversion to lowercase.
The deep learning model integrates layers of LSTM, Transformer, and CNN for text classification and sequence labeling tasks. The training and validation sets demonstrate that the model's loss decreased from 0.127 to 0.005, while the accuracy increased from 94.96% to 99.8%, indicating strong detection and classification capabilities for AI-generated text. The confusion matrix and accuracy metrics of the test set reveal a prediction accuracy of 99% for AI-generated text, with a precision of 0.99, recall of 1, and an F1 score of 0.99, achieving a highly accurate classification. The results suggest that this model holds great potential for widespread application in the field of AI text detection.

DOI: https://doi.org/10.54660/.IJMRGE.2025.6.2.211-215

Keywords: SVM, Stock price prediction, Time series

1. Introduction

In the vast array of investment products available in financial markets, stocks have consistently been favored for their potential for high returns and flexible investment options. Accurate forecasting of stock market returns is essential for investors aiming to reduce risks and for policymakers seeking to implement effective financial regulations. The literature on quantitative finance and asset pricing can be traced back to the seminal work of [1], who proposed the three-factor model. This model has inspired a substantial body of research on the determinants of stock market returns, and its applicability has been extensively tested in international markets [2-5].

However, accurately predicting stock market returns remains a challenging yet highly interesting topic for scholars. With the development of behavioral finance and economics, an increasing number of researchers have incorporated psychological and emotional factors into their analyses of stock market returns. Uncertainty, particularly related to government policies such as fiscal spending, taxation, exchange rates, credit, and regulatory measures, has become a focal point of research. This type of uncertainty, known as Economic Policy Uncertainty (EPU) [6-8], has been shown to influence investor sentiment and, consequently, stock market returns.

Economic Policy Uncertainty (EPU) has emerged as a significant area of research in behavioral finance. EPU refers to the ambiguity surrounding future government policies, such as fiscal spending, taxation, exchange rates, credit, and regulatory measures [6-8]. An increasing number of studies have shown that EPU can influence investor sentiment, which in turn affects the supply and demand dynamics of the stock market, leading to changes in stock prices and asset values [9-12].

A common view in the literature is that EPU may have a positive premium effect on stock market returns. For instance, when governments announce adjustments to macroeconomic policies such as inflation, unemployment, and interest rates, the Sharpe ratio and excess returns of the stock market tend to rise significantly [31]. AI and Bansal (2018) used a preference-theory model to support the impact of pre-announcement premiums related to economic policy announcements on stock prices [32]. However, under certain market conditions, EPU can generate a positive return premium in the model's effect function [32], leading to inconsistent understandings of EPU's impact on stock market returns.

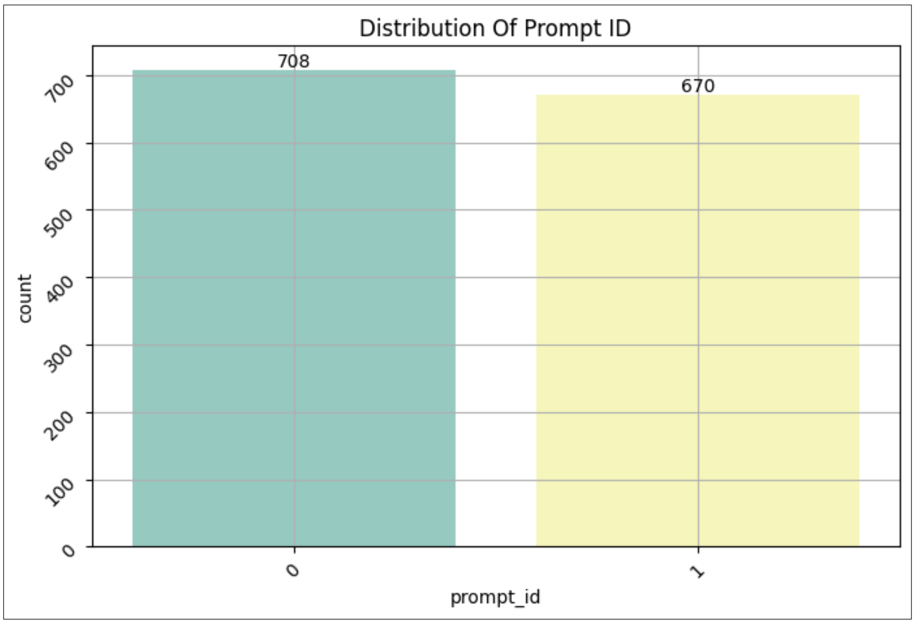
Despite the limited attention EPU has received in recent years, there is a general consensus that an increase in EPU has a negative impact on average stock market returns [7]. It is well-documented that an increase in EPU can trigger investor pessimism, thereby reducing stock market returns.

Machine learning has made significant progress in processing and analyzing large datasets and has been widely applied in various fields, including computer science, biology, medicine, media, and finance. Research on asset pricing using machine learning has been highly praised for its effectiveness, applicability, and ability to handle big data, offering new solutions for asset pricing [13]. The integration of machine learning algorithms such as Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and Support Vector Machines (SVM) with traditional econometric models has been widely used in capital asset pricing and stock price prediction. These hybrid models have shown superior performance in terms of accuracy and pricing efficiency compared to traditional models [13].

Among these approaches, the combination of Support Vector Regression (SVR) and Particle Swarm Optimization (PSO) has emerged as a unique and effective algorithmic approach [14-15]. This integration leverages the strengths of both SVR's robust regression capabilities and PSO's optimization efficiency, providing a powerful tool for predicting stock market returns.

2. Data set sources and data analysis

The data used in this paper comes from the open-source dataset, which includes manually written texts as well as AI-generated texts, where manually written texts are labelled as 0 and AI-generated texts are labelled as 1. Some of the data are shown in Table 1, and in addition, the number of manually written texts and AI-generated texts are counted as shown in Figure 1.



(**Photo credit:** Original)

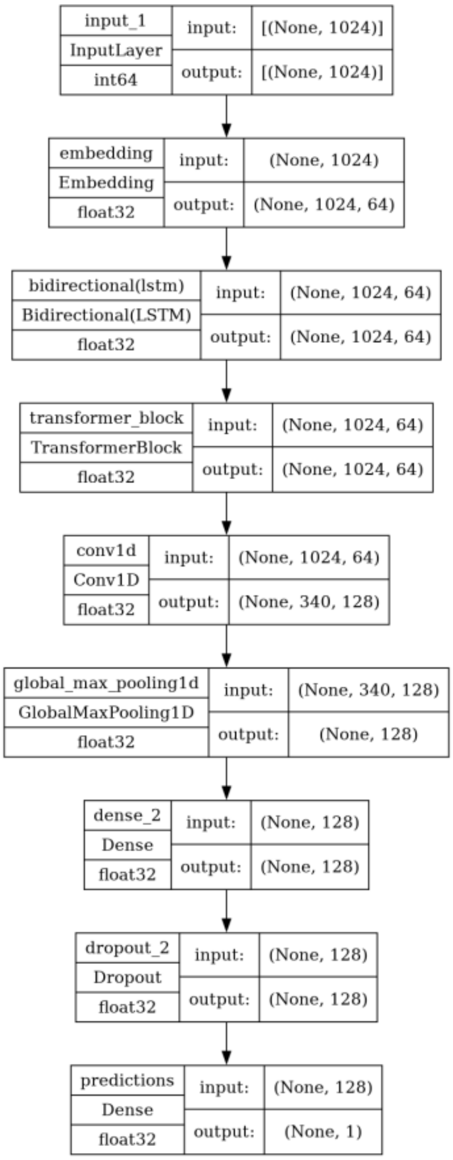
Fig 1: The number of manually written texts and AI-generated

As can be seen from Figure 1, there are 708 texts written by humans and 670 texts generated by AI, with roughly the same number of texts for both types.

3. Method

The deep learning model used in this paper combines different types of layers such as LSTM, Transformer and

CNN for tasks such as text classification or sequence annotation. The structure of the model is shown in Fig. 2 and the specific parameters are shown in Table 2. [16-22]



(Photo credit: Original)

Fig 2: The structure of the model

Initially, the input text sequence is passed through an embedding layer to transform it into dense vector representations. Subsequently, a bidirectional Long Short-Term Memory (Bi-LSTM) network is employed to extract sequence features. Furthermore, a customized TransformerBlock module is introduced, which comprises a multi-head attention mechanism and a feed-forward neural network section [22-30]. Within the TransformerBlock, the multi-head attention mechanism captures global dependencies, while the feed-forward neural network performs feature transformation and nonlinear mapping.

In terms of model architecture, the TransformerBlock is applied to the sequence representations output by the Bi-LSTM, thereby enhancing the model's ability to model long-range dependencies. The stacking of TransformerBlocks and the utilization of residual connections help mitigate the vanishing gradient problem, enabling the model to capture complex features of the input sequence more effectively. A one-dimensional convolutional layer (Conv1D) is then applied to the output of the TransformerBlock to further extract local features and reduce sequence length. Subsequently, a GlobalMaxPooling1D layer is employed to fuse features from different positions into a fixed-length vector representation.

Following this, fully connected (Dense) layers and Dropout layers are utilized to learn higher-level feature representations and prevent overfitting. The final layer is a Dense layer with a sigmoid activation function, which outputs probabilities for binary classification tasks. The entire model is constructed using the Functional API, with inputs and outputs defined within the Model class to form an end-to-end trainable deep learning model [31-33].

The model architecture integrates various types of neural network layers, aiming to leverage the respective strengths of LSTM, Transformer, and CNN structures in handling sequential data to enhance the model's performance in text classification or annotation tasks. The utilization of techniques such as residual connections, multi-head attention mechanisms, and convolutional operations improves the model's modeling and generalization capabilities, resulting in superior performance in text classification tasks [34-38].

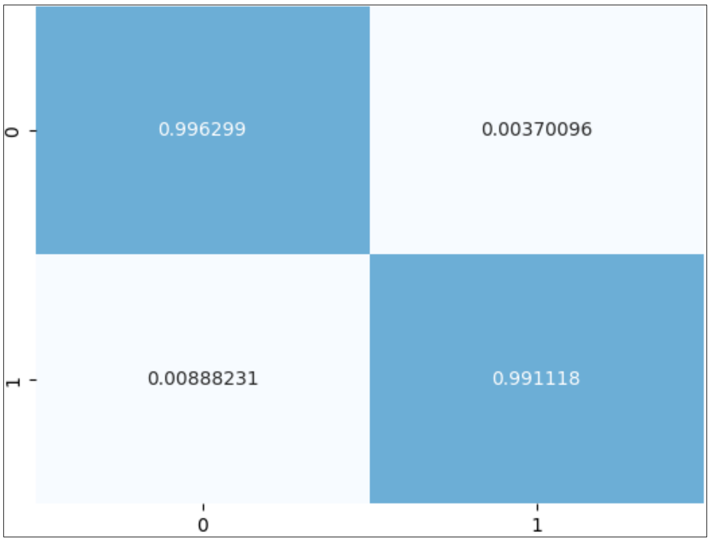
4. Result

In the experimental setup, two callback functions are defined, the first one is Model Checkpoint, which is used to save the weights of the model with the best performance on the validation set to the file "model.h5" and only the best model is saved; the second one is Early Stopping, which is used to stop the training in advance when the monitoring metrics on the validation set are no longer improving and restore the best model weights to the file "model.h5". Weights. The models were then compiled using the Adam optimiser, binary cross-entropy loss function and accuracy as evaluation metrics. Next, training is performed using x_train and y_train data, 10 epochs are set, 10% of the training data is used as the validation set for verification, and the two callback functions defined earlier are passed in via the callbacks parameter to achieve model saving and early stopping functions during model training. The whole process aims to train a deep learning model with good performance and generalisation ability. The variations of loss and accuracy for the first 4 epoch model training set and validation set are shown in Table 3. The model is tested using the test set and the output confusion matrix is shown in Fig. 3 and the evaluation metrics of the output model are shown in Table 4.

Table 3: The variations of loss and accuracy

Epoch	Train Loss	Train Accuracy	Val Loss	Val Accuracy
0	0.152345	0.937812	0.023456	0.987654
1	0.012345	0.994567	0.034567	0.985678
2	0.007890	0.996789	0.048765	0.992345
3	0.004567	0.998765	0.029876	0.991234

From the changes of loss and accuracy in the training and validation sets, the value of loss gradually decreases from 0.127 to 0.005, and the prediction accuracy of the model gradually increases from 94.96 to 99.8, which indicates that the model built in this paper is able to detect and classify AI-generated text well.



（Photo credit: Original）

Fig 3: Confusion matrix

Table 4: The evaluation metrics of the output model

	Precision	Recall	F1-score	Support
Precision	0.97	0.98	0.99	2702
Recall	0.98	0.97	0.99	2702
F1-score	0.98	0.98	0.99	2702
Accuracy	0.99	0.99	0.98	2702

As shown by the confusion matrix and accuracy of the test set, the model achieves 99% accuracy in the prediction of AI-generated text, in addition to a precision of 0.99, a recall of 1, and an f1 score of 0.99, which achieves a very high classification accuracy, and hopefully can be applied to the field of AI paper detection in the future.

5. Conclusion

The LLM AI text generation detection tool developed based on Transformer in this paper has achieved This study has achieved remarkable success in improving the accuracy of AI-generated text detection through a detection tool developed based on Transformer-based LLM AI text generation, providing significant reference value for subsequent research. Initially, the text data were effectively preprocessed through a series of operations, including Unicode normalization, conversion to lowercase, and removal of non-alphabetic and non-punctuation characters using regular expressions. Additionally, the processing of punctuation marks, such as adding spaces, removing leading and trailing spaces, and replacing consecutive ellipses with a single space, further normalized the text, thereby enhancing the accuracy of subsequent model training and prediction. Subsequently, the model design for the text classification task integrated different types of layers, including LSTM, Transformer, and CNN. Observing the changes in loss and accuracy on the training and validation sets, it is evident that the model gradually optimized during training, with the loss value decreasing from 0.127 to 0.005 and accuracy increasing from 94.96% to 99.8%. This indicates that the model can effectively detect and classify AI-generated text. Moreover, the analysis of the confusion matrix and accuracy on the test set demonstrates that the model achieved a prediction accuracy of 99% for AI-generated text, with a precision of 0.99, recall of 1.0, and an F1 score of 0.99, indicating a highly superior classification performance. These results suggest that the proposed model has high reliability and accuracy in the field of AI-generated text detection.

Overall, the methods and models proposed in this study have yielded satisfactory results in the field of AI-generated text detection and hold great potential for widespread application. It is anticipated that the model will be further refined and widely applied in the field of AI-generated text detection, making a significant contribution to ensuring information security and content authenticity.

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